

IMU

An IMU is a combination of sensors that measure orientation and motion with respect to an inertial reference frame.

Sensors in an IMU:

- 6 DOF IMU
Contains 3 axis gyroscope and 3 axis accelerometer
- 9 DOF IMU
Contains 3 axis gyroscope and 3 axis accelerometer and 3 axis magnetometer

Gyroscope: measure orientation angles and are angular rotation rate sensors. We integrate these rates to get angles but direct integration is not practical because of gyro bias and gyro drift. Even if the gyro is completely stationary on the desk it may display a rate that is constant with some irregularities and the constant is non zero this is known as gyro bias. When we integrate the rate wrt t and plot the graph it displays a linear graph with some irregularities and the angle is increasing this is known as gyro drift. Gyro bias causes Gyro drift. We can remove this using an extended kalman filter. The EKF will estimate the Gyro bias and then we correct the readings.

EKF: Linearizes the non-linear system model (kinematics) and measurement model (sensor readings) using Jacobian matrices (first-order Taylor series expansion).

Accelerometer: measure acceleration (linear, gravitational, centripetal, coriolis). Coriolis acceleration depends on the planet's rotation and our velocity (as it is $w \times v$). We can even measure our tilt angle using the accelerometers and we call these angles roll and pitch angle. Like the gyros the integration of the acceleration drifts away due to some biases and we can use EKFs for that. We can supplement the GPS readings to help correct those velocity and position estimates back to normal through sensor fusion and using EKFs.

Magnetometer: a digital compass that measures the strength and direction of magnetic fields, particularly Earth's magnetic field, to provide an absolute heading reference. This helps to determine orientation relative to magnetic north and corrects for drift in the gyroscope, improving the overall accuracy of motion tracking when combined with accelerometers and gyroscopes.

Sensor fusion: This is a process that combines data from multiple sensors to give us a more accurate, reliable and complete representation of a system.

- Accuracy:
By combining measurements of the same property so that noise and bias can be filtered out.
- Fault Tolerance:
Helps us when a sensor fails and there are other sensors that measure the same property that can take over and get the work done.

- Kalman Filters:
 Predict: Uses mathematical models of a system to reduce noise of a non-linear reading to linearize it
 Measure: Take data from all sensors to that measure the same property
 Update: The filter dynamically weighs the prediction against the measurement to adjust the uncertainty to the source of each measurement.

Orientation:

- Roll Pitch and Yaw
 Roll: angle about longitudinal axis
 Pitch: angle about lateral axis
 Yaw: angle about vertical axis
 Gimbal Lock: when 2 axis align eg. when pitch rotates +90 degrees the yaw and roll align
- Quaternions
 Quaternions are an extension of complex numbers that represent an axis-angle rotation in a compact, four-dimensional form. No gimbal lock. A unit quaternion q is represented by a scalar part and a 3D vector part:
 $q = w + xi + yj + zk$

IMU with ROS2:

1. Use a driver node to convert sensor data to ROS 2 messages
2. Identify the fields in these messages
3. Unit conversion if necessary
4. Publish the message to a topic
5. Define IMU's physical location and orientation relative to the rover's base
6. Perform Sensor Fusion

TYPES OF IMU

1. MEMS-based IMUs (Micro-Electro-Mechanical Systems)
 - Technology: These IMUs use tiny, silicon-based gyroscopes and accelerometers.
 - Characteristics: They are the smallest, lightest, and most cost-effective type. However, they also offer the lowest accuracy and exhibit the highest drift (the tendency for the calculated position/orientation to wander over time).
 - Applications: They dominate the consumer market, found in smartphones, smartwatches, small drones, and automotive safety systems.
2. FOG-based IMUs (Fiber Optic Gyroscope)
 - Technology: These devices use the Sagnac effect, passing light through long coils of optical fiber to measure rotation. They do not have moving parts.

- Characteristics: FOGs offer high accuracy and excellent stability over extended periods. They are significantly more expensive and larger than MEMS units. A key advantage is their immunity to electromagnetic interference.
- Applications: They are used in high-end robotics, tactical military systems, and maritime navigation where precise, reliable positioning is crucial.

3. RLG-based IMUs (Ring Laser Gyroscope)

- Technology: RLGs are the most precise inertial sensors. They use counter-propagating laser beams inside a rigid ring structure to detect rotational movement based on the interference pattern (Sagnac effect).
- Characteristics: These IMUs provide the highest possible precision and stability, with virtually no drift. They are also the largest and most expensive option.
- Applications: They are the standard for aerospace navigation, spacecraft guidance, and strategic defense applications.